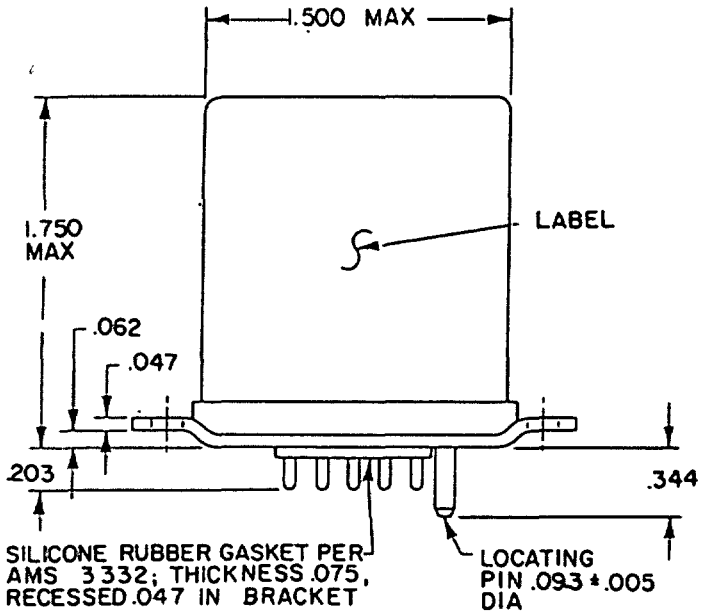
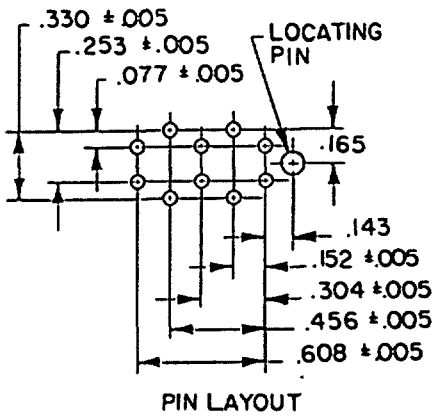
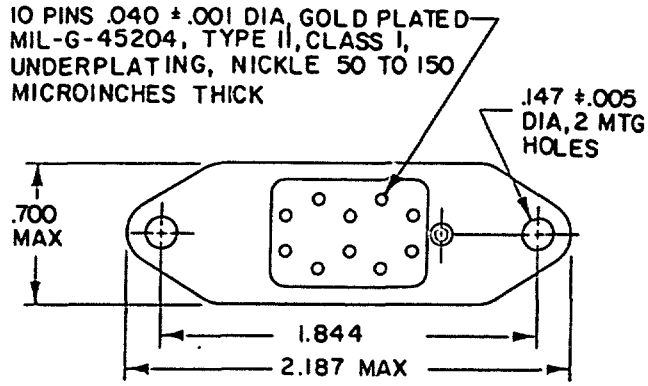
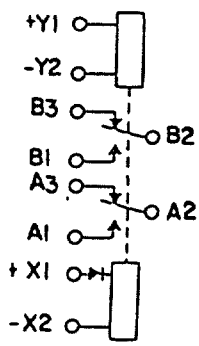
FED. SUP CLASS
5945

INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD
(FOR NEW DESIGN USE MIL-R-6106/38)

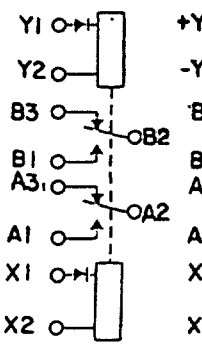


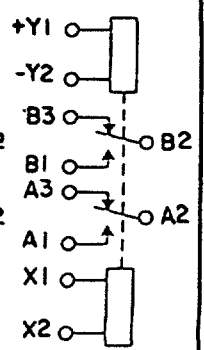




CIRCUIT DI
(NOTE 7)




CIRCUIT AI
(NOTE 5)




CIRCUIT ADI
(NOTES 5, 7)

Ⓕ ENTIRE STANDARD REVISED

APPROVED 1 Nov 60 REVISED Ⓕ 5 JUN 87

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 11

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

| P.A | International interest | TITLE | MILITARY STANDARD |
|---|---|---|---|
| AF - 85<br>Other Cust<br>Navy - AS | | RELAYS, ELECTROMAGNETIC, 5 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED | MS25455 |
| Procurement Specification<br>MIL-R-6106 | | SUPERSEDES: | PAGE 1 OF 5 |

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 5945-0745-15
AMSC N/A

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| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .165 | 4.19 |
| .005 | 0.13 | .203 | 5.16 |
| .010 | 0.25 | .253 | 6.43 |
| .040 | 1.02 | .304 | 7.72 |
| .047 | 1.19 | .330 | 8.38 |
| .062 | 1.57 | .344 | 8.74 |
| .075 | 1.91 | .456 | 11.58 |
| .077 | 1.96 | .608 | 15.44 |
| .093 | 2.36 | .700 | 17.78 |
| .143 | 3.63 | 1.500 | 38.10 |
| .147 | 3.73 | 1.750 | 44.45 |
| .152 | 3.86 | 1.844 | 46.84 |
| | | 2.187 | 55.55 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Pins to be perpendicular to header surface within one degree.
7. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
8. Shock, vibration, and acceleration requirements applicable with coils de-energized.
9. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
10. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

| Dash number MS25455- | Type | Coil | Terminal type | Max weight in pounds |
|---|---|---|---|---|
| D1 | I | dc | Plug in | 0.21 |
| A1 | I | ac | Plug in | 0.23 |
| AD1 | I | ac-dc | Plug in | 0.23 |

APPROVED 1 Nov 1960 REVISED (F) ENTIRE STANDARD REVISED.

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11
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Review activities: Army -
Navy - EC
Air Force - 11

User activities: Army -
Navy -
Air Force -

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Procurement Specification: MIL-R-6106

P.A: AF - 85
Other Cust: Navy - AS

International Interest:

SUPERSEDES:

TITLE: RELAYS, ELECTROMAGNETIC, 5 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25455

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FED. SUP CLASS 5945

TABLE II. Operating characteristics.

| MS part no. MS25455- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Operate 3/ | Release 4/ | Contact bounce | | | | |
| | | Volts 1/ | Freq. Hz | Res Ω | Volts | Amperes | Normal 2/ | High temp test | Cont current test | | | Main | | Aux | | |
| | | | | | | | | | | | | NO | NC | NO | NC | |
| D1 | X1, X2 Y1, Y2 | 28 | dc | N/A | 29 | 0.12 | 18 | 18 | 19.8 | 25 | N/A | 2 | 2 | | | |
| A1 | X1, X2 Y1, Y2 | 115 | 400 5/ | N/A | 122 | 0.06 | 90 | 90 | 95 | 25 | N/A | 2 | 2 | | | |
| AD1 | X1, X2 | 115 | 400 5/ | N/A | 122 | 0.06 | 90 | 90 | 95 | 25 | N/A | 2 | 2 | | | |
| | Y1, Y2 | 28 | dc | N/A | 29 | 0.12 | 18 | 18 | 19.8 | 25 | N/A | 2 | 2 | | | |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25455-A1 and -AD1 ac coils may be used on 60 Hz if maximum ambient temperature is limited to +85°C (coil current: 0.066 ampere maximum).

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Review activities: Army -
Navy - EC
Air Force - 11

User activities: Army -
Navy -
Air Force -

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Procurement Specification MIL-R-6106

P.A AF - 85
Other Cust Navy - AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 5 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25455

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FED. SUP CLASS 5945

TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operating cycles X $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 5 | 5 | | | 5 | 4 | | | | | | | |
| Inductive | 100 | | | | | | | | | | | | | |
| Inductive | 20 | 3 | 3 | | | 3 | 2 | | | | | | | |
| Motor | 100 | 1.5 | 1.5 | | | 1.5 | 1 | | | | | | | |
| Lamp | 100 | 0.8 | 0.8 | | | 0.8 | 0.6 | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 1.25 | 1.25 | | | 1.25 | 1 | | | | | | | |
| Intmd current | Applicable per specification | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Review activities: Army -
Navy - EC
Air Force - 11

User activities: Army -
Navy -
Air Force -

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Procurement Specification MIL-R-6106

P.A AF - 85
Other Cust Navy - AS

International interest

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 5 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25455

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FED. SUP CLASS 5945

## Environmental characteristics

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 G |
| Duration | 11 ms |
| Max duration contact opening | 10 µs |
| Vibration - sinusoidal | |
| G-level | 10 G |
| Frequency range | 5 - 1500 Hz |
| Acceleration | 15 G |

## Electrical characteristics

| | |
|---|---|
| Insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |

Dielectric strength (sea level)

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | | |
| All other points | 1,000 V rms | 1,000 V rms |

| Dielectric strength (altitude) | (When mounted in mating socket) 80,000 ft |
|---|---|
| Coil to case | 500 V rms |
| Aux contacts | |
| All other points | 500 V rms |

| | |
|---|---|
| Max contact drop initial | 0.150 volt |
| After life test | 0.175 volt |
| Overload current | 20 amperes |
| Rupture current | 25 amperes |
| Duty rating | Continuous |
| RFI specification | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays)

## Quality conformance inspection

Performance of groups B and C inspections are not applicable.

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